

Chapter 2

First Order Differential Equation

(Follows from Dennis G. Zill [1])

2.2 Separable Variables

$$\frac{dy}{dx} = f(x, y) \quad (2.1)$$

Definition 2.2.1 Separable Equation

$$\frac{dy}{dx} = g(x)h(y) \quad (2.2)$$

For example,

$$(2.3) \quad \frac{dy}{dx} = x^2 y \text{ is a separable equation and}$$

can be solved by separable variables method:

$$\frac{dy}{y} = x^2 dx \Rightarrow \int \frac{dy}{y} = \int x^2 dx \Rightarrow \ln y = \frac{x^3}{3} + \ln c$$

$$\Rightarrow \ln y - \ln c = \frac{x^3}{3} \Rightarrow \ln \frac{y}{c} = \frac{x^3}{3}$$

$$\Rightarrow \frac{y}{c} = e^{\frac{x^3}{3}} \Rightarrow y = ce^{\frac{x^3}{3}} \quad \square$$

⊗ But (2.4) $\frac{dy}{dx} = x^2 + y$ is a non separable equation,

and cannot be solved by the above method.

Example 1 Solve $(1 + x)dy - ydx = 0$

$$\Rightarrow (1 + x)dy = ydx \quad \Rightarrow \frac{dy}{y} = \frac{dx}{1 + x}$$

$$\Rightarrow \int \frac{dy}{y} = \int \frac{dx}{1 + x} \quad \Rightarrow \ln|y| = \ln|1 + x| + \ln c_1$$

$$\ln \frac{|y|}{c_1} = \ln|1 + x| \quad \Rightarrow \frac{|y|}{c_1} = |1 + x|$$

$$y = \pm c_1(1 + x) \quad \Rightarrow \quad y = c(1 + x) \quad \square$$

Example 2 Example 4

Exercise 2.2

(Zill) 1 – 7, 15 – 17, 23 – 26.

1. $\frac{dy}{dx} = \sin 5x$

$$\Rightarrow dy = \sin 5x \, dx \quad \Rightarrow \int dy = \int \sin 5x \, dx$$

$$\Rightarrow y = -\frac{\cos 5x}{5} + c \quad \square$$

$$5. \quad x \frac{dy}{dx} = 4y$$

$$\Rightarrow \frac{dy}{y} = 4 \frac{dx}{x} \quad \Rightarrow \int \frac{dy}{y} = 4 \int \frac{dx}{x}$$

$$\Rightarrow \ln|y| = 4 \ln|x| + \ln c_1 \quad \Rightarrow \ln \frac{|y|}{c_1} = \ln|x|^4$$

$$\Rightarrow |y| = c_1 |x|^4 \quad \Rightarrow y = \pm c_1 x^4 \quad \Rightarrow y = c x^4 \quad \square$$

$$(\text{IVP}) \quad 23. \quad \frac{dx}{dt} = 4(x^2 + 1), \quad x\left(\frac{\pi}{4}\right) = 1$$

$$\Rightarrow \frac{dx}{1+x^2} = 4dt \quad \Rightarrow \int \frac{dx}{1+x^2} = \int 4dt$$

$$\Rightarrow \tan^{-1} x = 4t + c \quad \dots (i) \quad \Rightarrow x = \tan(4t + c) \quad \dots (ii)$$

$$\text{Using i.c. } x\left(\frac{\pi}{4}\right) = 1 \text{ in } (i), \quad \tan^{-1} 1 = 4\frac{\pi}{4} + c$$

$$\Rightarrow \frac{\pi}{4} = \pi + c \quad \Rightarrow c = -3\frac{\pi}{4}$$

$$\text{Put the value of } c \text{ in } (ii), \quad x = \tan\left(4t - \frac{3\pi}{4}\right) \quad \square$$

$$(IVP) \quad 25. \quad x^2 \frac{dy}{dx} = y - xy, \quad y(-1) = -1$$

$$\Rightarrow x^2 \frac{dy}{dx} = y(1 - x) \quad \Rightarrow \frac{dy}{y} = \frac{1 - x}{x^2} dx$$

$$\int \frac{dy}{y} = \int \frac{1 - x}{x^2} dx \Rightarrow \int \frac{dy}{y} = \int \left(\frac{1}{x^2} - \frac{1}{x} \right) dx$$

$$\Rightarrow \ln y = -\frac{1}{x} - \ln x + \ln c \Rightarrow \ln \frac{xy}{c} = -\frac{1}{x}$$

$$\frac{xy}{c} = e^{-\frac{1}{x}} \Rightarrow y = c \frac{1}{x} e^{-\frac{1}{x}} \quad (i)$$

$$\text{Using I.C., } -1 = c(-1)e \Rightarrow 1 = ce \Rightarrow c = e^{-1}$$

$$\text{Now put the value of } c \text{ in (i), } y = \frac{1}{x} e^{-\left(\frac{1}{x} + 1\right)} \quad \square$$

2.3 Linear Equations

Definition 2.3.1 Linear Equations

$$a_1(x) \frac{dy}{dx} + a_0(x)y = g(x) \quad (1)$$

Standard Form

$$\frac{dy}{dx} + P(x)y = f(x) \quad (2)$$

Solution method for (1)

1. Write (1) into the standard form (2).
2. From (2), find integrating factor $e^{\int p(x)dx} = I(x)$.
3. Multiply both sides of (2) by $I(x)$

$$\Rightarrow \frac{d}{dx} [I(x)y] = I(x)f(x)$$

4. Integrate the above equation: $\int d[I(x)y] = \int I(x)f(x) dx$
5. Solve for y .

Example 1 – 3, 5

Exercise 2.3 1 – 10, 25 – 27

Exercise 2.3 Solution

$$3. \frac{dy}{dx} + y = e^{3x} \quad (1)$$

Here, $p(x) = 1$, I.F. $e^{\int p(x)dx} = e^{\int dx} = e^x$.

Multiply (1) by I.F. e^x

$$e^x \frac{dy}{dx} + e^x y = e^x e^{3x} \Rightarrow \frac{d}{dx} [ye^x] = e^{4x}$$

$$\Rightarrow \int d[ye^x] = \int e^{4x} dx \Rightarrow ye^x = \frac{1}{4} e^{4x} + c$$

$$\Rightarrow y = \frac{1}{4} e^{3x} + ce^{-x}; \quad x \in (-\infty, +\infty) \quad \square$$

$$9. \quad x \frac{dy}{dx} - y = x^2 \sin x \quad (1)$$

$$\Rightarrow \text{S.F.} \quad \frac{dy}{dx} - \frac{y}{x} = x \sin x \quad (2)$$

$$p(x) = -\frac{1}{x}, \text{ I.F. } e^{\int p(x)dx} = e^{\int -\frac{1}{x} dx} = e^{-\ln x} = e^{\ln x^{-1}} = \frac{1}{x}$$

Multiply (2) by I.F. $1/x$,

$$\frac{1}{x} \frac{dy}{dx} - \frac{y}{x^2} = \sin x \Rightarrow \frac{d}{dx} \left[y \frac{1}{x} \right] = \sin x$$

$$\Rightarrow \int d \left[y \frac{1}{x} \right] = \int \sin x dx \Rightarrow y \frac{1}{x} = -\cos x + c$$

$$\Rightarrow y = -x \cos x + cx; \quad x \in (0, \infty) \quad \square$$

27. (IVP) $xy' + y = e^x; \quad y(1) = 2$

S. F. DE $\frac{dy}{dx} + \frac{y}{x} = \frac{e^x}{x}$ (1)

Here, $p(x) = \frac{1}{x}$,

$\therefore$ I. F. $e^{\int p(x)dx} = e^{\int \frac{1}{x}dx} = e^{\ln x} = x$

Multiply (1) by I.F. x ,

$$x \frac{dy}{dx} + y = e^x \Rightarrow \frac{d}{dx} [yx] = e^x$$

$$\int d[yx] = \int e^x dx \Rightarrow yx = e^x + c$$

$$\Rightarrow y = \frac{e^x}{x} + \frac{c}{x} \quad (2)$$

Now using the I.V. $y(1) = e + c = 2 \Rightarrow c = 2 - e$

Putting the value of c into (2),

$$y = \frac{e^x}{x} + \frac{2 - e}{x} \quad ; \quad x \in (0, \infty) \quad \square$$

2.4 Exact Equations

Examples

$$ydx + xdy = 0; \quad x^2y^3dx + x^3y^2dy = 0$$

are exact equations, since the left hand side expressions are exact differentials, i.e., since,

$$d(xy) = ydx + xdy; \quad d\left(\frac{1}{3}x^3y^3\right) = x^2y^3dx + x^3y^2dy$$

Definition 2.4.0 Exact Differential

A differential expression $M(x, y)dx + N(x, y)dy$ is an exact differential in a region R in the xy -plane, if the expression can be expressed as a total differential of some function $f(x, y)$ defined in R , i.e. if,

$$df(x, y) = M(x, y)dx + N(x, y)dy.$$

Definition 2.4.1 Exact Equation

A first order differential equation of the form

$$M(x, y)dx + N(x, y)dy = 0$$

is said to be an exact equation, if the expression on the left-hand side is an exact differential.

Theorem 2.4.1 Criterion for an Exact Differential

A necessary and sufficient condition that $M(x, y)dx + N(x, y)dy$ be an exact differential is

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

Example 1, 2, 4.

Exercise 2.4 1 – 5, 31 – 33

Solution 2.4

$$3. (5x + 4y)dx + (4x - 8y^3)dy = 0 \quad (0)$$

$$M(x, y) = 5x + 4y, \quad N(x, y) = 4x - 8y^3$$

$$\text{Here, } \frac{\partial M}{\partial y} = 4 = \frac{\partial N}{\partial x}$$

Therefore, equation (0) is an exact equation.

$$\text{So, } df(x, y) = M(x, y)dx + N(x, y)dy \quad (1)$$

$$\Rightarrow \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy = M(x, y)dx + N(x, y)dy$$

$$\Rightarrow \frac{\partial f}{\partial x} = M(x, y) = 5x + 4y \quad (2)$$

$$\Rightarrow \frac{\partial f}{\partial y} = N(x, y) = 4x - 8y^3 \quad (3)$$

Integrating (2) partially w.r.to x , treating y as constant,

$$f(x, y) = \int (5x + 4y) dx = \frac{5}{2} x^2 + 4xy + g(y) \quad (4)$$

Differentiating (4) partially, w.r.to y ,

$$\frac{\partial f}{\partial y} = 4x + g'(y) = N(x, y) = 4x - 8y^3 \quad (5)$$

$$\Rightarrow g'(y) = -8y^3 \Rightarrow g(y) = -2y^4$$

Putting the value of $g(y)$ into (4),

$$f(x, y) = \frac{5}{2} x^2 + 4xy - 2y^4 \quad (6)$$

Since from (1),

$$df(x, y) = 0 \Rightarrow f(x, y) = c \text{ (const.)},$$

Therefore, by (6), the solution of the DE (0) is obtained in implicit form as

$$\frac{5}{2} x^2 + 4xy - 2y^4 = c \quad \square$$

$$33. \quad 6xy \, dx + (4y + 9x^2)dy = 0 \quad (1)$$

$$M_y = 6x, N_x = 18x, \Rightarrow p(y) = \frac{N_x - M_y}{M} = \frac{12x}{6xy} = \frac{2}{y}$$

$$\therefore \text{I.F. } e^{\int p(y)dy} = e^{\int \frac{2}{y} dy} = e^{2 \ln y} = e^{\ln y^2} = y^2$$

Multiply (1) by I.F. y^2 ,

$$Pdx + Qdy \equiv 6xy^3 \, dx + (4y^3 + 9x^2y^2)dy = 0 \quad (2)$$

$$P_y = 18xy^2 = Q_x$$

Therefore, equation (2) is an exact equation.

$$\text{So, } df(x, y) = P(x, y)dx + Q(x, y)dy \quad (3)$$

$$\Rightarrow \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy = P(x, y)dx + Q(x, y)dy$$

$$\Rightarrow \frac{\partial f}{\partial x} = P(x, y) = 6xy^3 \quad (4)$$

$$\Rightarrow \frac{\partial f}{\partial y} = Q(x, y) = 4y^3 + 9x^2y^2 \quad (5)$$

Integrating (4) partially w.r.to x , treating y as constant,

$$f(x, y) = \int 6xy^3 \, dx = 3x^2y^3 + g(y) \quad (6)$$

Differentiating (6) partially, w.r.to y ,

$$\frac{\partial f}{\partial y} = 9x^2y^2 + g'(y) = Q(x, y) = 4y^3 + 9x^2y^2 \quad (7)$$

$$\Rightarrow g'(y) = 4y^3 \Rightarrow g(y) = y^4$$

Putting the value of $g(y)$ into (6),

$$f(x, y) = 3x^2y^3 + y^4 \quad (8)$$

Since from (2) & (3),

$$df(x, y) = 0 \Rightarrow f(x, y) = c \text{ (const.)},$$

Therefore, by (6), the solution of the DE (1) is obtained in implicit form as

$$3x^2y^3 + y^4 = c \quad \square$$

2.5 Solutions by Substitutions

Bernoulli's Equation The differential equation

$$\frac{dy}{dx} + P(x)y = f(x)y^n, \quad (1)$$

where n is any real number, is called **Bernoulli's equation**. Note that for $n = 0$ and $n = 1$, equation (1) is linear. For $n \neq 0$ and $n \neq 1$ the substitution $u = y^{1-n}$ reduces any equation of the form to a linear equation.

Example 2.

2.5 Exercise (15 – 25)

2.5 Solutions

$$15. \quad x \frac{dy}{dx} + y = \frac{1}{y^2} \quad (1) \Rightarrow \text{B.E.} \quad \frac{dy}{dx} + \frac{y}{x} = \frac{y^{-2}}{x} \quad (2)$$

Here $n = -2$, $\therefore$ we substitute, $u = y^{1-(-2)} = y^3$

$$\Rightarrow y = u^{\frac{1}{3}} \text{ and } \frac{dy}{dx} = \frac{d}{du} \left[u^{\frac{1}{3}} \right] \frac{du}{dx} = \frac{1}{3} u^{-\frac{2}{3}} \frac{du}{dx} \quad (3)$$

Using (3) into (2),

$$\frac{1}{3} u^{-\frac{2}{3}} \frac{du}{dx} + \frac{u^{\frac{1}{3}}}{x} = \frac{u^{-\frac{2}{3}}}{x} \Rightarrow \frac{du}{dx} + \frac{3u}{x} = \frac{3}{x} \quad (4)$$

Now to solve the S.F. linear DE (4), we find,

$$\text{I.F. } e^{\int p(x)dx} = e^{\int \frac{3}{x} dx} = e^{3 \ln x} = e^{\ln x^3} = x^3$$

Multiply (4) by I.F. x^3 ,

$$x^3 \frac{du}{dx} + 3x^2 u = 3x^2 \quad \Rightarrow \frac{d}{dx} [ux^3] = 3x^2$$

$$\int d[ux^3] = \int 3x^2 dx \quad \Rightarrow ux^3 = x^3 + c$$

$$\Rightarrow u = 1 + \frac{c}{x^3} \quad \Rightarrow y^3 = 1 + \frac{c}{x^3} \quad \square$$

$$18. \quad x \frac{dy}{dx} - (1+x)y = xy^2 \quad (1)$$

$$\Rightarrow \text{B.E.} \quad \frac{dy}{dx} - \left(\frac{1}{x} + 1\right)y = y^2 \quad (2)$$

Here $n = 2$, $\therefore$ we substitute, $u = y^{1-(2)} = \frac{1}{y}$

$$\Rightarrow y = \frac{1}{u} \quad \text{and} \quad \frac{dy}{dx} = -\frac{1}{u^2} \frac{du}{dx} \quad (3)$$

Using (3) into (2),

$$\begin{aligned} & -\frac{1}{u^2} \frac{du}{dx} - \left(\frac{1}{x} + 1\right) \frac{1}{u} = \frac{1}{u^2} \\ \Rightarrow & \frac{du}{dx} + \left(\frac{1}{x} + 1\right)u = -1 \end{aligned} \quad (4)$$

$$\text{I.F. } e^{\int p(x)dx} = e^{\int (\frac{1}{x}+1)dx} = e^{\ln x + x} = e^{\ln x} e^x = x e^x$$

Multiply (4) by $x e^x$,

$$x e^x \frac{du}{dx} + (e^x + x e^x)u = -x e^x$$

$$\Rightarrow \frac{d}{dx} [u x e^x] = -x e^x$$

$$\int d[u x e^x] = \int -x e^x dx \Rightarrow u x e^x = -(x e^x - e^x) + c$$

$$u = \left(\frac{1}{x} - 1 \right) + \frac{c}{x e^x} \Rightarrow \frac{1}{y} = \frac{1}{x} - 1 + \frac{c}{x e^x} \quad \square$$

1. A First Course in Differential Equations with Modeling and Applications, (10th Edition), Author-Dennis G. Zill.